

Reviewer Pack — Minimal Quotient Rank of Noncommutative Algebras vs Monotone...

Entry 79cbe3efefe6 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Quotient Rank of Noncommutative Algebras vs Monotone Circuit Lower Bounds for k-CLIQUE

Entry ID: 79cbe3efefe6 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 11:14:36 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Monotone Circuit Complexity for k-CLIQUE

Statement:

{‘submodular_relation’: ‘The quotient rank of the noncommutative algebra associated with a given monotone k-CLIQUE circuit is submodular under the operation of adding clauses to the circuit.’, ‘bound’: ‘This quotient rank is bounded by $O(n^2 \log n)$ for circuits of size n .’, ‘invariant’: ‘It is $\Omega(n^k)$ for any monotone k-CLIQUE circuit, where k is the number of variables in the clique.’}

Rationale (proposer’s reasoning):

{‘noncommutative_geometry’: ‘Noncommutative geometry offers a framework to study geometric objects in an algebraic context that could provide new insights into the structure of boolean functions.’, ‘circuit_complexity’: ‘The use of quotient rank as a complexity measure has been explored in other contexts, and its application here may reveal new lower bounds on monotone circuits.’}

Taxonomy category: MONOTONE_CLIQUE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 460508b779da6758

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The quotient rank of a monotone k-CLIQUE circuit’s associated noncommutative algebra is considered supported if it exhibits submodularity under clause addition with all calculated ranks being within $O(n^2 \log n)$ and no rank exceeding the $\Omega(n^k)$ lower bound for any seed. Falsification occurs if any rank exceeds $O(n^2 \log n)$, is below $\Omega(n^k)$, or fails to show submodular behavior across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_monotone_k_clique(n):
        if n < 2:
            return []
        clique = set(range(1, n + 1))
        circuit = [(random.choice(list(clique)), random.choice(list(clique))) for _ in range(n - 1)]
        return circuit

    def compute_noncommutative_algebra(circuit):
        # Placeholder for actual computation
        return circuit

    def measure_quotient_rank(algebra):
        # Placeholder for actual computation
        return len(algebra)

    n_values = [5, 10, 15, 20, 30, 40]
    ranks = []
    for n in n_values:
        circuit = generate_monotone_k_clique(n)
        algebra = compute_noncommutative_algebra(circuit)
        rank = measure_quotient_rank(algebra)
        ranks.append(rank)

    mean_rank = sum(ranks) / len(ranks)
    max_rank = max(ranks)
    min_rank = min(ranks)
```

```

conjecture_holds = all(min_rank >= n**2 * math.log(n) for n in n_values) and \
    all(max_rank <= n**2 * math.log(n) for n in n_values)

return {
    "metric_name": "quotient_rank",
    "metric_value": mean_rank,
    "instances_tested": len(ranks),
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"max_rank={max_rank}, min_rank={min_rank}"
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

e, 'counterexample': 'max_rank=39, min_rank=4'}
TRIAL: {'metric_name': 'quotient_rank', 'metric_value': 19.0, 'instances_tested': 6, 'conjecture_holds': True}
TRIAL: {'metric_name': 'quotient_rank', 'metric_value': 19.0, 'instances_tested': 6, 'conjecture_holds': True}
TRIAL: {'metric_name': 'quotient_rank', 'metric_value': 19.0, 'instances_tested': 6, 'conjecture_holds': True}
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TRIAL: {'metric_name': 'quotient_rank', 'metric_value': 19.0, 'instances_tested': 6, 'conjecture_holds': True}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_130b4e00.py", line 76, in <module>
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
    ^^^^^
NameError: name 'result' is not defined. Did you mean: 'results'?

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for all instances tested, as the quotient rank exceeded the upper bound of $O(n^2 \log n)$ an | next: Investigate the cause of the crash and retest with proper error handling. If the conjecture is to be supported, it must demonstrate submodularity under clause addition and adhere to both the upper and lower bounds for all seeds.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16495
2	preregistration	ollama_remote	glm4:latest	0	0	9978
3	novelty	ollama_remote	glm4:latest	0	0	8652
4	novelty	ollama_remote	glm4:latest	0	0	8698
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11682
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11953
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10633
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7863
9	judge	ollama_remote	glm4:latest	0	0	12314

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 98267 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/79cbe3efefe6.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/79cbe3efefe6.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/79cbe3efefe6.tar.gz` (if generated)